

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 September 2026
Team ID	
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Login	Registration through Service Portal form (name, mobile, email, password) Registration through Gmail Login with email / mobile and password
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Journey Selection & Fare Calculation	Select source station and destination station Select number of passengers and ticket type (single / round-trip) Automatic fare calculation from the fare matrix, including round-trip discount
FR-4	Ticket Booking	Ticket request through ServiceNow Service Catalog item Booking record creation in ServiceNow Booking summary and confirmation before payment
FR-5	Digital Payment (Simulated)	Payment mode selection: UPI, credit / debit card, mobile wallet (simulated gateway) Payment status (success / failure) with transaction reference number Payment failure handling and retry
FR-6	QR Code Ticket Generation	Unique QR code generated automatically after successful payment (Flow Designer) Ticket validity period and single-use rule QR ticket displayed in Service Portal and sent by email / SMS
FR-7	Ticket Validation (AFC)	QR validation at entry and exit through the Gate Validation page Ticket status update (Issued / Used / Expired) Rejection of invalid, duplicate or expired QR codes
FR-8	Ticket Management	View booking history Download ticket Cancel unused ticket and refund as per policy

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-9	Notifications	Booking confirmation and payment receipt Ticket expiry reminder (email / SMS)
FR-10	Station Manager Dashboard & Reports	Daily sales reconciliation Ticket sales per station / route and payment mode Peak-hour and passenger analytics
FR-11	Administration	Manage stations, routes and fare matrix Manage user roles and access Audit logs of bookings and payments

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple three-step booking (select route, pay, receive QR) on mobile and desktop browsers; responsive ServiceNow Service Portal; clear error and status messages so that first-time commuters can book without assistance.
NFR-2	Security	Role-based access control for Commuter, Customer Care Executive, Gate Operator, Station Manager and Administrator through ServiceNow roles and ACLs; HTTPS / TLS for data in transit; passwords stored as hashes; no real card or UPI credentials stored (payment is simulated and only a transaction reference is kept); unique, tamper-resistant QR codes to prevent forgery or reuse.
NFR-3	Reliability	No booking is lost if a payment fails or the session drops; pending payments are retried and reconciled; every booking, payment and ticket status change is traceable through audit logs.
NFR-4	Performance	QR ticket generated within 5 seconds of payment confirmation; QR validation at the gate completes within 2 seconds; the system supports multiple concurrent commuters during peak hours.
NFR-5	Availability	Service available 24x7 on the ServiceNow cloud instance with a target uptime of 99.5% or higher; issued QR tickets remain accessible to the commuter after booking.
NFR-6	Scalability	New stations, routes, fares and user roles are added through configuration rather than code changes; the platform scales to a growing number of commuters and daily transactions.